CLASS 9 — SCIENCE FULL SYLLABUS PAPER SET 01

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The speed of light is now defined to be exactly: 1

(a) 186000 m/s only **(b)** 300 m/s **(c)** 299792458 m/s **(d)** 3 km/s

2. For measuring the time to cycle from school to home, a useful simple model would keep: 1

(a) the school's name **(b)** distance and average speed

(c) the colour of the cycle **(d)** the make of the tyres

3. Smoke is an example of a heterogeneous mixture of: 1

(a) solid particles in a gas **(b)** two gases

(c) a liquid in a solid **(d)** a gas in a liquid

4. Muddy water is an example of a: 1

(a) suspension **(b)** pure substance **(c)** colloid **(d)** true solution

5. Which compound is ionic? 1

(a) $MgCl_2$ **(b)** CO_2 **(c)** HCl **(d)** CH_4

6. Sugar solution does not conduct electricity because: 1

(a) sugar is a metal **(b)** water is a conductor only

(c) sugar is covalent and does not form ions **(d)** sugar forms Na^+ ions

7. The electronic configuration (K, L, M, N shells) of the sodium atom (Z = 11) is: 1

(a) 8, 2, 1 **(b)** 2, 8, 2 **(c)** 2, 8, 1 **(d)** 2, 9

8. A vehicle moving at 30 m s^{-1} is brought to rest with a uniform retardation of 5 m s^{-2}. The stopping distance is: 1

(a) 45 m **(b)** 180 m **(c)** 90 m **(d)** 6 m

9. Which graph shows a body moving with uniform acceleration? 1

(a) a horizontal line on a velocity–time graph

(b) a straight inclined line on a velocity–time graph

(c) a vertical line on a position–time graph

(d) a horizontal line on a position–time graph

10. The net force needed to give a body of mass 2 kg an acceleration of 6 m s^{-2} is: 1

(a) 8 N **(b)** 12 N **(c)** 24 N **(d)** 0.33 N

11. The weight of a 60 kg body is (g = 10 m s^{-2}): 1

(a) 60 N **(b)** 1200 N **(c)** 600 N **(d)** 300 N

12. If the velocity of a vehicle doubles, its kinetic energy becomes: 1

(a) four times **(b)** half **(c)** same **(d)** two times

13. Pitch of a sound depends mainly on its: 1

(a) intensity **(b)** amplitude **(c)** frequency **(d)** speed

14. The lifespan of a red blood cell is about: 1

(a) 10 years **(b)** 12 days **(c)** 1 year **(d)** 120 days

15. Uncontrolled cell division that forms tumours is characteristic of: 1

(a) gametes **(b)** red blood cells **(c)** cancer cells **(d)** root hair cells

16. F. C. Steward grew whole carrot plants from single: 1

(a) phloem cells **(b)** xylem cells **(c)** guard cells **(d)** root hair cells

17. The correct order of taxonomic hierarchy from the highest is: 1

(a) Kingdom, Class, Phylum, Order, Genus, Family, Species
(b) Kingdom, Phylum, Class, Order, Family, Genus, Species
(c) Phylum, Kingdom, Class, Family, Order, Genus, Species
(d) Species, Genus, Family, Order, Class, Phylum, Kingdom

18. Sexual reproduction produces variations because: 1

(a) cells divide by budding **(b)** spores are formed
(c) there is only one parent **(d)** gametes from two parents combine

19. Gametes are formed by: 1

(a) mitosis **(b)** budding **(c)** meiosis **(d)** fragmentation

20. Air moves from: 1

(a) high pressure to low pressure **(b)** cold to cold
(c) low pressure to high pressure **(d)** sea to sea

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Why is estimation a useful scientific skill? 2

22. Define a molecule. Give one example each of a molecule of an element and of a compound. 2

23. A car starting from rest has acceleration 3 m s^{-2}. Find its velocity after 4 s. 2

24. A bullet of mass 20 g is fired from a gun. Why is the recoil of the gun much smaller in velocity than the bullet's velocity? 2

25. What is the difference between mitosis and meiosis in one line? 2

26. Differentiate between a tendon and a ligament. 2

27. What is pollination? Name two agents. 2

28. Name the processes of the water cycle. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Why does a torch beam passing through a glass of milk-water mixture look visible while it is invisible through a glass of salt solution? 3

30. Distinguish between isotopes and isobars with one example each. 3

31. A crane lifts a mass to the 10th floor in time t, then to the 20th floor in 2t. Compare the work and power. 3

32. State three applications of ultrasound. 3

33. Compare bone and cartilage. 3

34. State two features each of birds and mammals. 3

35. What are the energy changes when atoms combine? Why are molecules more stable than separate atoms? 3

OR

Calculate the molecular mass of H_2SO_4 and the formula unit mass of $CaCl_2$ (H=1, S=32, O=16, Ca=40, Cl=35.5). 3

36. How does the oceanic reservoir regulate carbon in the atmosphere? 3

OR

What is the Haber–Bosch process and its impact? 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Show that the combining ratio of carbon and oxygen in CO_2 is 3 : 8. 5
(ii) How much carbon is in 22 g of CO_2?
(iii) Find the mass of oxygen needed to burn 6 g of carbon completely.

OR

Case study: A student reacted zinc with dilute hydrochloric acid in an open test tube and found the mass decreased. 5
(i) Name the products.
(ii) Why did the mass decrease?
(iii) How can she modify the set-up to verify the conservation of mass?
(iv) What result should she get?

38. **Case study:** A car of 800 kg moving at 25 m s^{-1} stops in 10 s. 5
(i) Find the retardation.
(ii) Find the braking force.
(iii) Find the stopping distance.
(iv) If the speed were doubled for the same force, find the new stopping distance.

OR

(i) State Newton's second law in terms of momentum. 5
(ii) Define momentum.
(iii) A 2 kg ball moving at 5 m s^{-1} is stopped in 0.5 s; find the force.

39. (i) What is the cell cycle? 5
(ii) Compare mitosis and meiosis.
(iii) State the importance of cell division.

OR

Case study: In cell culture, cells are grown in a nutrient medium at a fixed temperature. 5

(i) What is cell culture?

(ii) Why must conditions be sterile?

(iii) Name one use.

(iv) What is totipotency?

40. (i) Compare external and internal fertilisation. 5

(ii) Why is the survival rate higher in internal fertilisation?

(iii) What is IVF?

OR

Case study: ASHA workers advise mothers on care during pregnancy and after birth. 5

(i) Name one thing she suggests for the mother's diet.

(ii) Why is breast-feeding recommended?

(iii) What is post-partum depression?

(iv) Why should hygiene be maintained?

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND